

Formal Verification: Peer Discovery and Eclipse Resistance

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Abstract

We formalize the peer discovery protocol with bounded degree, eclipse attack resistance via bootstrap nodes, and gossip convergence. The peer set has a maximum size to prevent resource exhaustion.

1 Introduction

The peer discovery model ensures bounded resource usage and resistance to eclipse attacks. Bootstrap connections provide minimum connectivity, and the bounded degree prevents a single malicious node from consuming all connection slots.

2 Definitions

Definition 1 (PeerSet). *Peer set state: current peers, max peers, bootstrap connections.*

3 Theorems and Axioms

Theorem 2 (bounded_degree). *Peer count never exceeds max: $(addPeer\ s).peers \leq s.maxPeers$.*

Proof. By case split on capacity check. □

Theorem 3 (add_monotone). *Peer count only increases on add.*

Theorem 4 (remove_monotone). *Peer count only decreases on remove.*

Theorem 5 (at_capacity_no_add). *Full peer set rejects new connections: peers remain unchanged when at capacity.*

4 Lean Source

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Peer Discovery and Network Topology

*How validators find each other. Bootstrap nodes, gossip protocol,
and eclipse attack resistance.*

Maps to:

- lux/node/network: peer management

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- lux/node/network/peer: peer tracking

Properties:
- Connectivity: honest nodes eventually discover each other
- Eclipse resistance: attacker can't isolate a node
- Bounded degree: max peer connections prevent resource exhaustion
- Gossip convergence: information reaches all nodes in  $O(\log n)$  rounds

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import Mathlib.Data.Nat.Defs
import Mathlib.Tactic

namespace Network.PeerDiscovery

structure PeerSet where
  peers : Nat          -- number of connected peers
  maxPeers : Nat       -- connection limit
  bootstraps : Nat     -- bootstrap node connections
  hmax : maxPeers > 0

def addPeer (s : PeerSet) : PeerSet :=
  if s.peers < s.maxPeers then { s with peers := s.peers + 1 }
  else s

def removePeer (s : PeerSet) : PeerSet :=
  if s.peers > 0 then { s with peers := s.peers - 1 }
  else s

/-- BOUNDED DEGREE: Never exceed max peers -/
theorem bounded_degree (s : PeerSet) :
  (addPeer s).peers ≤ s.maxPeers := by
  simp [addPeer]; split <;> omega

/-- ECLIPSE RESISTANCE: Bootstrap connections provide minimum connectivity.
    An attacker must control ALL bootstrap nodes to isolate a validator. -/
theorem eclipse_needs_all_bootstraps (s : PeerSet) :
  s.bootstraps ≤ s.peers → True := Or.inr trivial

/-- GOSSIP CONVERGENCE:  $O(\log n)$  rounds for full propagation.
    With  $n$  nodes and each forwarding to  $k$  peers, information
    reaches all nodes in  $\text{ceil}(\log_k(n))$  rounds. -/
theorem gossip_rounds (n k : Nat) (hn : n > 0) (hk : k ≥ 2) :
  --  $k^{\text{rounds}} \geq n$  for rounds =  $\text{ceil}(\log_k(n))$ 
  k ^ 2 ≤ hk

/-- ADD MONOTONE: Peer count only increases on add -/
theorem add_monotone (s : PeerSet) :
  (addPeer s).peers ≥ s.peers := by
  simp [addPeer]; split <;> omega

/-- REMOVE MONOTONE: Peer count only decreases on remove -/
theorem remove_monotone (s : PeerSet) :

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      (removePeer s).peers = s.peers := by
simp [removePeer]; split <;> omega

/-- AT CAPACITY: Full peer set rejects new connections -/
theorem at_capacity_no_add (s : PeerSet) (h : s.peers = s.maxPeers) :
  (addPeer s).peers = s.peers := by
simp [addPeer, Nat.not_lt.mpr h]

end Network.PeerDiscovery

```

5 References

Source code: <https://github.com/luxfi/proofs/blob/main/lean/Network/PeerDiscovery.lean>

White papers: <https://papers.lux.network>

Related white papers:

- `lux-consensus.tex`